

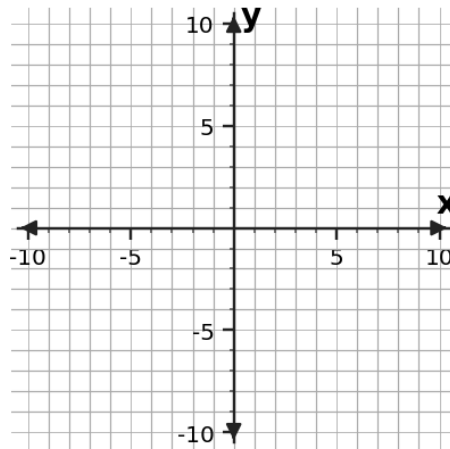
Unit 1 • Targeted Practice • 24 Questions

Show enough work to make your thinking clear. Use the graph or table when one is supplied.

I Can: Plot and read ordered pairs.

1. Plot a Given Ordered Pair

Plot the point T(-8,4) on the coordinate plane. Label the point T.



Answer: T is at (-8, 4).

Solution: Move 8 units left from the origin and 4 units up; place and label T.

2. Translate Movement to an Ordered Pair

Start at the origin. Move 9 units up, then 4 units left. Write the ordered pair for the ending point.

Answer: (-4, 9)

Solution: Left makes the x-coordinate -4. Up makes the y-coordinate 9.

3. Diagnose a Coordinate Sign Error

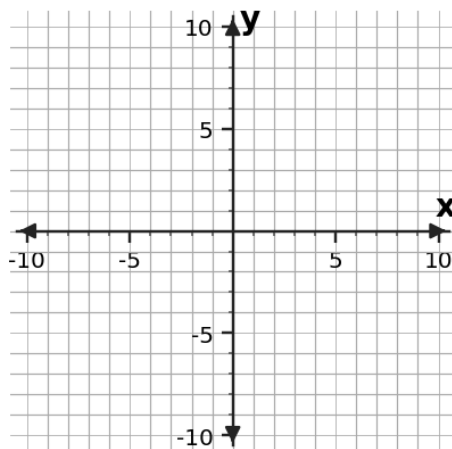
A student says, "From the origin I moved 7 units right and 4 units down, so the point is (-7,-4)." Explain the error and write the correct ordered pair.

Answer: The x-coordinate should be positive. The point is (7, -4).

Solution: Moving right gives a positive x-coordinate; moving down gives a negative y-coordinate.

4. Plot a Given Ordered Pair

Plot the point $U(6,0)$ on the coordinate plane. Label the point U .



Answer: U is at $(6, 0)$ on the x -axis.

Solution: Move 6 units right and 0 units vertically; place U on the x -axis.

5. Translate Movement to an Ordered Pair

Start at the origin. Move 5 units up, then 9 units right. Write the ordered pair for the ending point.

Answer: $(9, 5)$

Solution: Right gives $x=9$; up gives $y=5$.

6. Diagnose a Coordinate Sign Error

A student says, “Two units left and 6 units down ends at $(-2,6)$.” Explain the error and write the correct ordered pair.

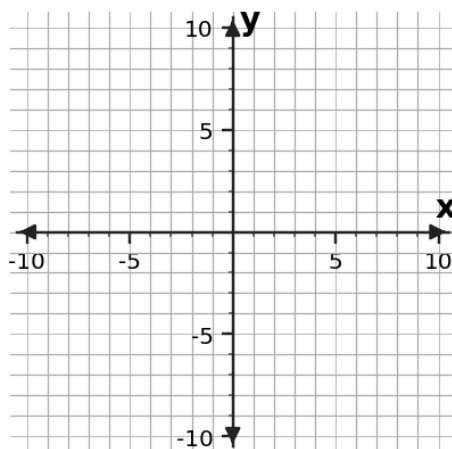
Answer: The y -coordinate should be negative. The point is $(-2, -6)$.

Solution: Left gives a negative x -coordinate and down gives a negative y -coordinate.

I Can: Use the x- and y-axes correctly.

7. Axis-First Coordinate Move

For the point $V(-7,3)$, state which axis controls each movement. Then plot and label V.

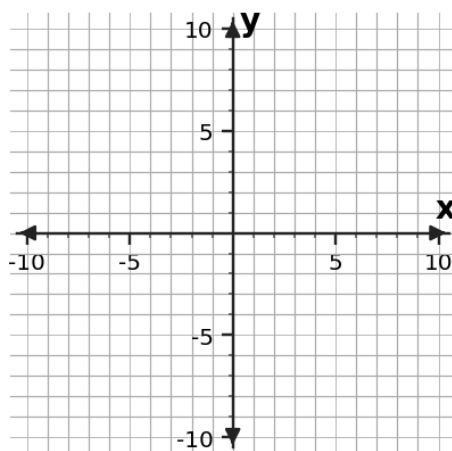


Answer: x-axis: 7 left; y-axis: 3 up; $V(-7,3)$.

Solution: The first coordinate controls horizontal x movement; the second controls vertical y movement.

8. Assign Variables to Axes

A graph will show the number of photos uploaded as time passes. Which quantity belongs on the x-axis, and which belongs on the y-axis? Explain why.



Answer: Time on the x-axis; number of photos uploaded on the y-axis.

Solution: Time is the input that progresses; the upload count is measured for each time.

9. Diagnose an Axis Swap

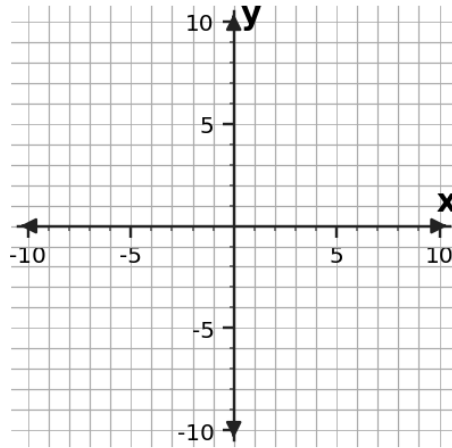
To plot $W(4,-8)$, Jordan moves 8 units right and 4 units down. What did Jordan reverse? State the correct movements.

Answer: Jordan reversed the coordinates. Move 4 right, then 8 down.

Solution: The first coordinate, 4, controls x; the second, -8, controls y.

10. Axis-First Coordinate Move

For the point $X(0,-9)$, describe the x-movement and y-movement. Then plot and label X .



Answer: x-movement: 0; y-movement: 9 down; $X=(0,-9)$.

Solution: Because $x=0$ there is no horizontal movement. The y-coordinate -9 places X on the negative y-axis.

11. Assign Variables to Axes

A student wants to graph the temperature of a cooling drink as time passes. The student puts temperature on the x-axis and time on the y-axis. Is that assignment sensible? If not, correct it and explain.

Answer: No. Time should be on the x-axis and temperature on the y-axis.

Solution: Time is the input/independent quantity; temperature is recorded as the output for each time.

12. Diagnose an Axis Swap

To plot $Y(-5,2)$, Alex moves 5 units up and then 2 units left. Explain the axis mistake and state the correct movements.

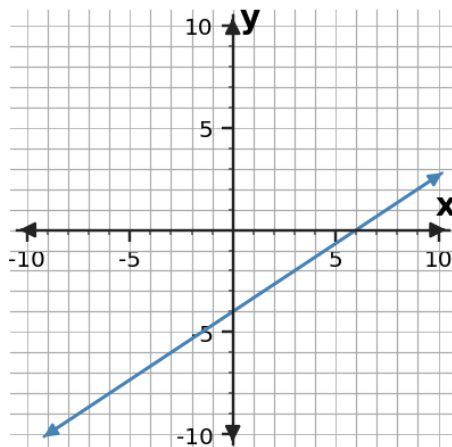
Answer: Alex swapped the axis roles. Move 5 left, then 2 up.

Solution: The x-coordinate -5 controls horizontal movement; $y=2$ controls vertical movement.

I Can: Identify and interpret intercepts when they are shown.

13. Identify Both Intercepts

Identify the x-intercept and y-intercept of the line.

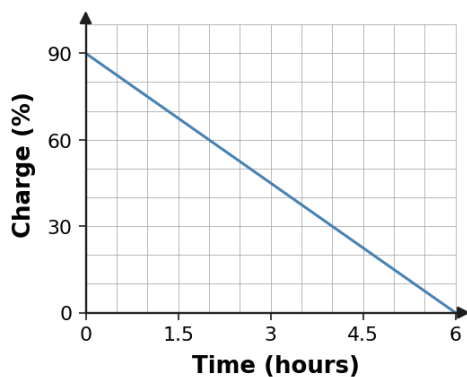


Answer: x-intercept (6,0); y-intercept (0,-4).

Solution: At the x-intercept $y=0$; at the y-intercept $x=0$.

14. Interpret an Intercept in Context

The graph shows the percent charge remaining in a device as streaming time passes. What does the y-intercept mean in this situation? Include units.

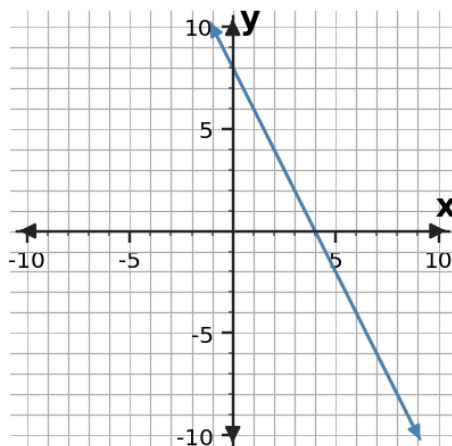


Answer: At 0 hours, the device has 90% charge.

Solution: The y-intercept occurs when time is 0, so it represents the starting charge: 90%.

15. Distinguish x- from y-Intercept

A student says, “(4,0) is the y-intercept because it lies on an axis.” Correct the statement and explain how you know.

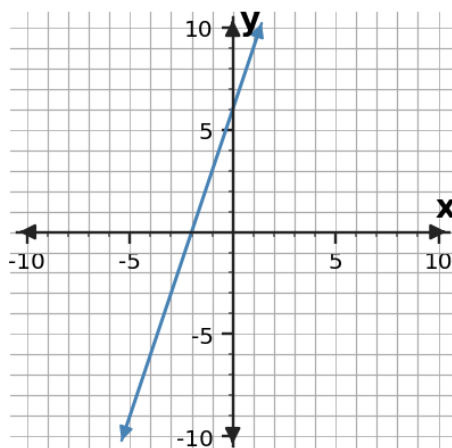


Answer: (4,0) is the x-intercept; the y-intercept is (0,8).

Solution: An x-intercept has $y=0$. A y-intercept has $x=0$.

16. Identify Both Intercepts

Identify the x-intercept and y-intercept of the line.

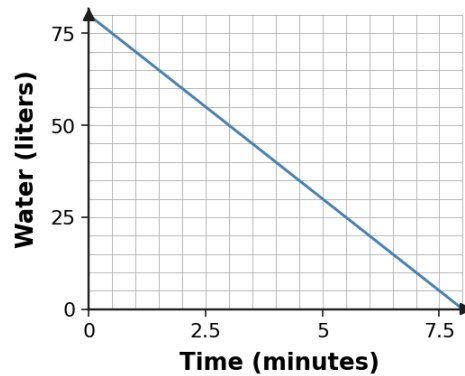


Answer: x-intercept (-2,0); y-intercept (0,6).

Solution: The x-intercept is where $y=0$; the y-intercept is where $x=0$.

17. Interpret an Intercept in Context

The graph shows the amount of water left in a tank as it drains. What does the x-intercept mean in this situation? Include units.

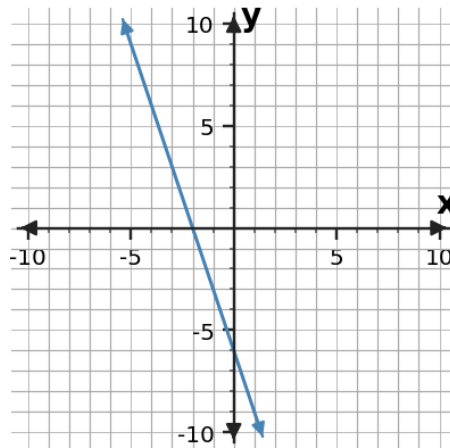


Answer: The tank is empty after 8 minutes.

Solution: The x-intercept is $(8, 0)$, meaning 0 liters remain at 8 minutes.

18. Distinguish x- from y-Intercept

A student calls $(0, -6)$ the x-intercept. Correct the statement, identify both intercepts, and give a rule that distinguishes them.



Answer: $(0, -6)$ is the y-intercept; the x-intercept is $(-2, 0)$.

Solution: A y-intercept has $x=0$. An x-intercept has $y=0$.

I Can: Find and interpret rate of change from a context.

19. Compute and Interpret Context Rate

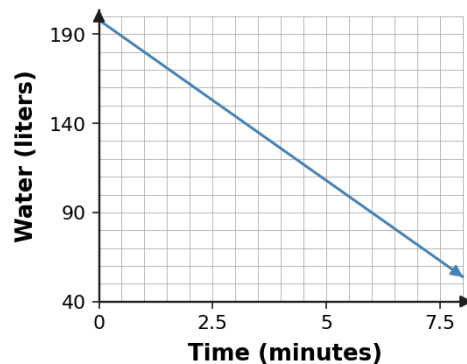
A tank contains 120 liters of water after 3 minutes and 192 liters after 7 minutes. Find the rate of change and interpret it in context.

Answer: 18 liters per minute; the amount of water increases by 18 liters each minute.

Solution: Rate = $(192-120)/(7-3) = 72/4 = 18$ liters per minute.

20. Read Rate from a Context Graph

The graph shows water remaining in a reservoir as time passes. Find the rate of change using two readable points and explain what the sign means.



Answer: -18 liters per minute; the reservoir loses 18 liters each minute.

Solution: For example, from (1,180) to (5,108): $(108-180)/(5-1) = -72/4 = -18$ L/min. The negative sign means the amount decreases.

21. Distinguish Rate from Initial Value

A drone's height is modeled by $H=250-8t$, where H is height in feet and t is time in seconds. A student says, "The rate is 250 feet per second." Explain the error and state the actual rate.

Answer: The rate is -8 feet per second; 250 feet is the initial height.

Solution: The coefficient of t gives change per second. The constant is the value at $t=0$.

22. Compute and Interpret Context Rate

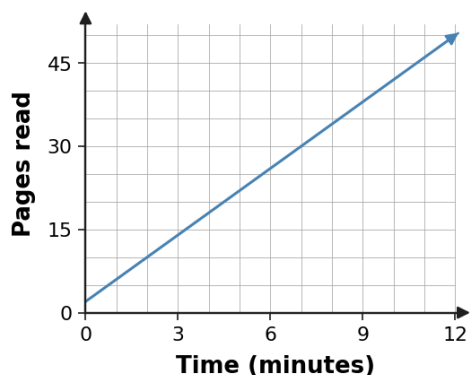
A cup of tea is 82°F after 1 minute and 61°F after 8 minutes. Find the rate of change and interpret it in context.

Answer: -3°F per minute; the temperature decreases by 3°F each minute.

Solution: Rate = $(61-82)/(8-1) = -21/7 = -3^\circ\text{F}$ per minute.

23. Read Rate from a Context Graph

The graph shows the number of pages read as time passes. Find the rate of change and explain what it means.



Answer: 4 pages per minute; the reader gains 4 pages for each additional minute.

Solution: Using (4,18) and (10,42): $(42-18)/(10-4) = 24/6 = 4$ pages per minute.

24. Distinguish Rate from Initial Value

A scooter rental app gives the table below. A student says, “The rate is \$4 per minute because \$4 is the first cost shown.” Explain the error and state the actual rate.

Minutes	Total cost
0	\$4.00
5	\$11.50
10	\$19.00

Answer: The rate is \$1.50 per minute; \$4 is the starting fee.

Solution: The cost rises \$7.50 over 5 minutes, so rate = $7.50/5 = \$1.50$ per minute. At 0 minutes the cost is \$4, so \$4 is the initial value.